Extraction and Comparison of Anthocyanin Rich Plant Dyes as Cleaner Solar Cell Photosynthesizers

INTRODUCTION

Over 60% of American energy is sourced from fossil fuels (Transformative Power Systems, Energy.gov) and ~9% from renewable energy. Solar panels have evolved through three generations. The first is the silicon based solar cell, toxic in production due to carcinogen release during silicon processing (Is Silicone an Eco-Friendly Material?. 2020). Minimal research exists on the Third Generation Solar Cell, the Dye Sensitized Solar Cell (DSSC), which utilizes chemical or natural dye as the photosynthesizer.

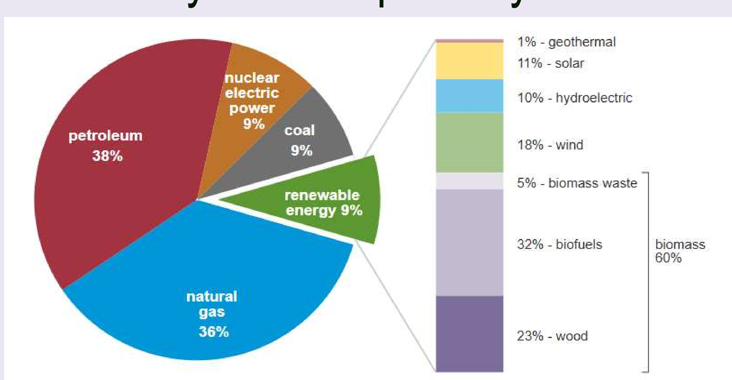

Fig.1: U.S. Energy Sourcing (U. S. Energy Facts Explained - Consumption and Production - u. S. Energy Information Administration(Eia), n.d.)

Purple/blue plants get their unique hue from Anthocyanins, a naturally occurring chemical whose molecular structure absorbs visible light (Shukor et al., 2023). So, dye from anthocyanin rich plants can act as a natural photosynthesizer in DSSCs.

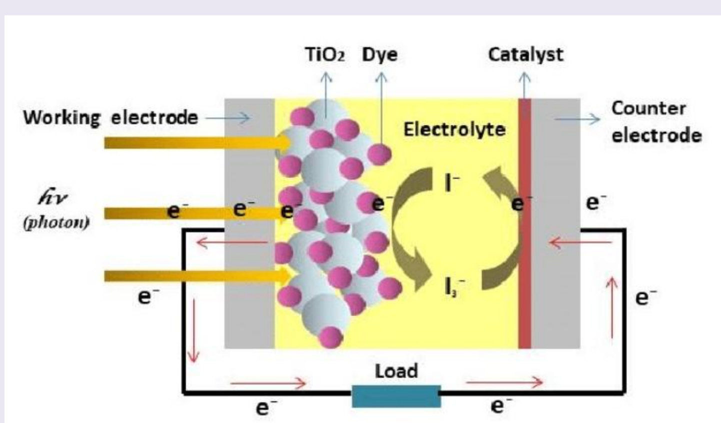

Fig.2: Jamalullail, N., Ili Salwani Mohamad, Mohd Natashah Norizan, N. A. Baharum, & Norsuria Mahmed. (2017, December). *Short Review: Natural pigments photosynthesizers for dye-sensitized solar cell (DSSC)* [Conference]. https://doi.org/10.1109/SCORED.2017.8305367

DSSCs have four main components:

Working Electrode: A thin coat of semiconducting material with wide energy band gaps acting as a porous wall to which the dye molecules bond that initializes electron transfer.

Photosynthesizer: Light reactive dye with a molecular structure that absorbs visible light that excites upon photon incident, resulting in electron ejection (oxidizing the dye).

Electrolyte: A chemical compound made of ionic liquids and cations, possessing a redox couple. It "recharges" oxidized dye by "lending" its extra electron. The electron is regained when the circuit electrons reach the anode.

Counter Electrode: A conductive material that catalyzes the reduction of the electrolyte after electron transfer.

PURPOSE

Utilizing natural plant dyes as photosynthesizers is a new development in scientific research, so the methodology of creating DSSCs with plant dyes is still ambiguous. The first step to clarifying this process lies in determining what kind of plant dye photosynthesizer produces the highest energy efficiency. Once a reliable methodology for creating plant-based DSSCs is created, multiple cells may be connected in a series circuit to power small devices such as water heaters, LEDs, or small fans.

METHODS

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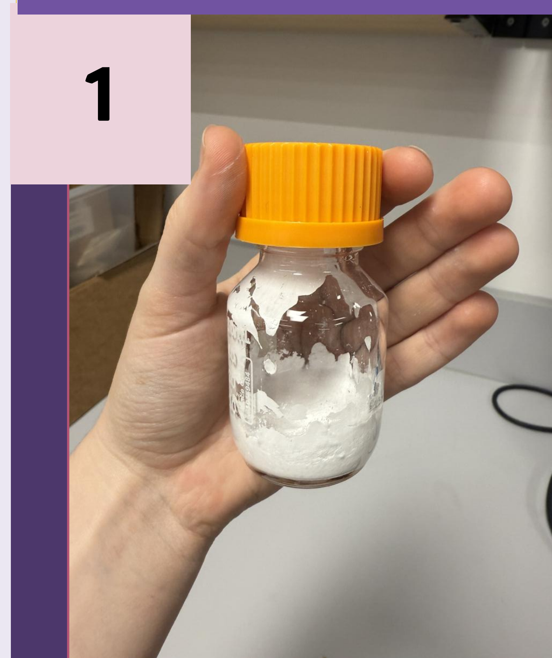

Fig.3: TiO2 Paste (Photo Taken by Finalist, 2025)

1) Chemical Prep.

TiO2 paste is created for the working electrode and Lugol's Iodine (I3-) as the liquid electrolyte.

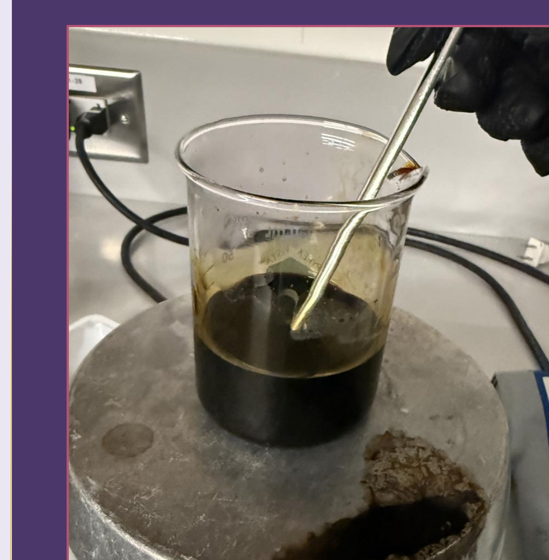

Fig.4: Lugol's Iodine (Photo Taken by Finalist, 2025)

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Fig.5: Cabbage Dye Extraction (Photo Taken by Finalist, 2025)

2) Dye Extraction

The plant is cut into small pieces and heated in 100mL of absolute ethanol until 75mL of solution remains.

Fig.6: Dye Extraction (Photo Taken by Finalist, 2026)

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Fig.7: Annealing Kiln (Photo Taken by Finalist, 2025)

3) Cell Assembly

ITO Glass plates are coated in TiO2 Paste & graphite, annealed, then soaked with dye and Lugol's Iodine.

Fig. 8: Dye Coating (Photo Taken by Finalist, 2026)

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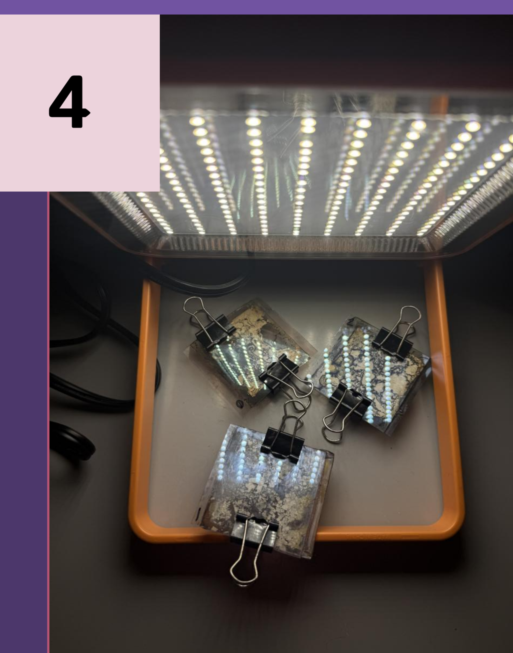


Fig.9: Cell Charging (Photo Taken by Finalist, 2026)

4) Testing

The cells are charged under a halogen lamp then tested with a multimeter for voltage, resistance, and amperage.

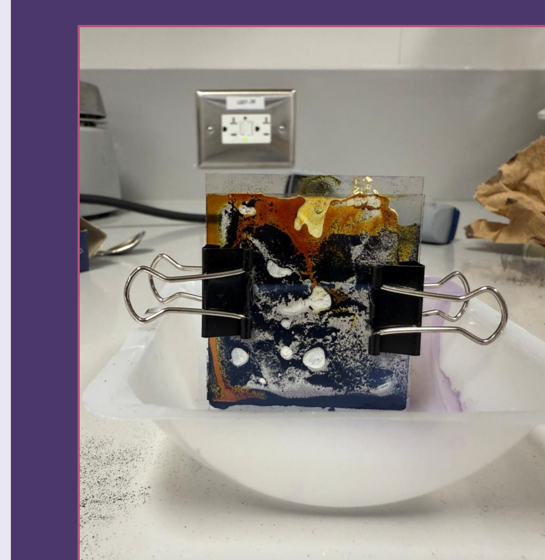
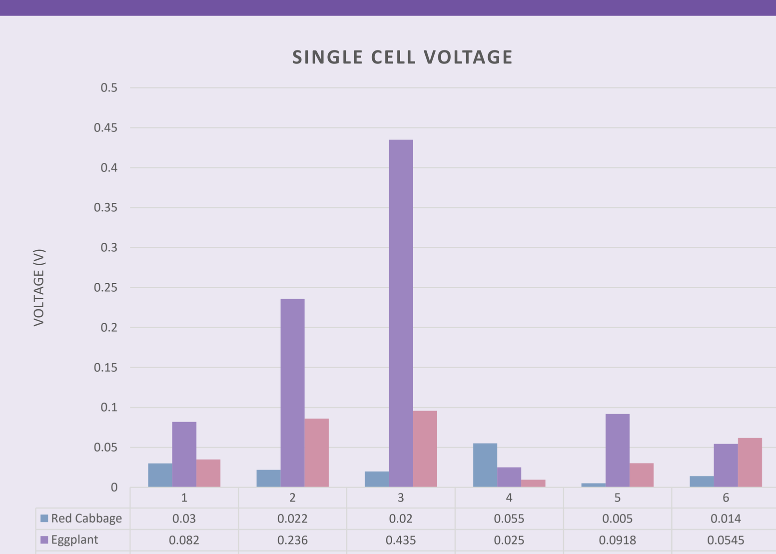


Fig. 10: Fresh Cell (Photo Taken by Finalist, 2026)

RESULTS



	1	2	3	4	5	6
Red Cabbage	0.03	0.022	0.02	0.055	0.005	0.014
Eggplant	0.082	0.236	0.435	0.025	0.0918	0.0545
Purple Potato	0.035	0.086	0.0958	0.0095	0.0301	0.0618

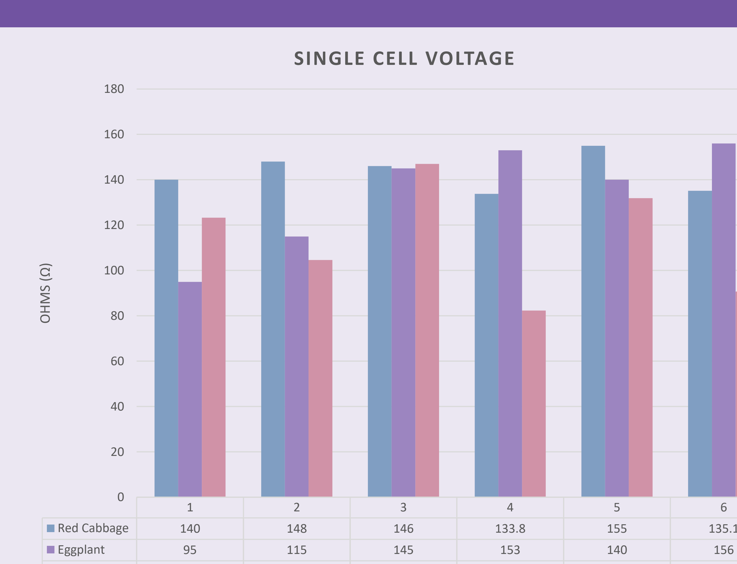
Fig.11 Voltage by Test (Graph Created by Finalist Using Excel, 2026)

Voltage Comparison

Red Cabbage Cells
Highest Voltage: 0.055V
Lowest Voltage: 0.005V

Eggplant Skin Cells
Highest Voltage: 0.435V
Lowest Voltage: 0.082V

Purple Potato Cells
Highest Voltage: 0.0958V
Lowest Voltage: 0.0095V



	1	2	3	4	5	6
Red Cabbage	140	148	146	133.8	155	135.1
Eggplant	95	115	145	153	140	156
Purple Potato	123.3	104.6	147	82.3	131.9	90.7

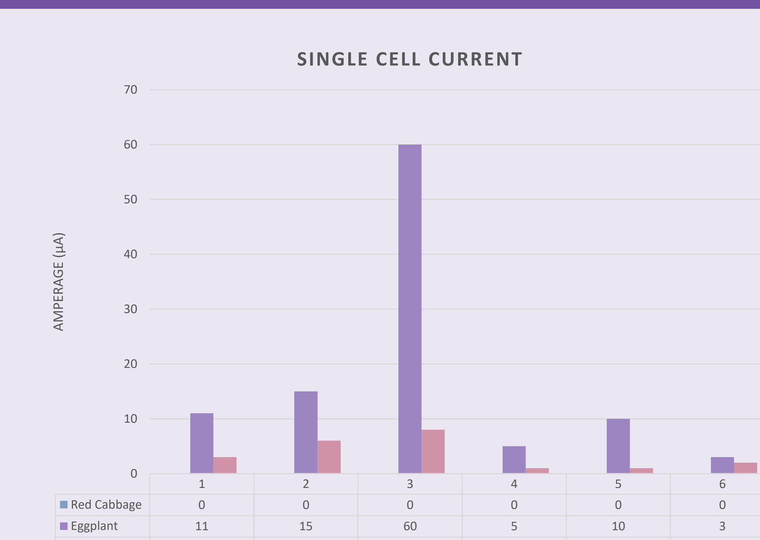
Fig.12 Resistance by Test (Graph Created by Finalist Using Excel, 2026)

Resistance Comparison

Red Cabbage Cells
Lowest Resistance: 133.8 Ω
Highest Resistance: 155.0 Ω

Eggplant Skin Cells
Lowest Resistance: 95.0 Ω
Highest Resistance: 145.0 Ω

Purple Potato Cells
Lowest Resistance: 82.3 Ω
Highest Resistance: 147.0 Ω



	1	2	3	4	5	6
Red Cabbage	0	0	0	0	0	0
Eggplant	11	15	60	5	10	3
Purple Potato	3	6	8	1	1	2

Fig 13: Amperage by Test (Graph Created by Finalist Using Excel, 2026)

Amperage Comparison

Red Cabbage Cells
Highest Current: 0μA
Lowest Current: 0μA

Eggplant Skin Cells
Highest Current: 60μA
Lowest Current: 11μA

Purple Potato Cells
Highest Current: 8μA
Lowest Current: 1μA

CONCLUSIONS

The plant dye that produced the highest voltage was the eggplant skin dye with a maximum reading of 0.435V. While the red cabbage cells did not produce amperage, the eggplant cells produced up to 60μA and potato cells up to 8 μA. Further experimentation is required to fully understand the differences that led to increased amperage, but the current data suggests that it may be due to the different type of dye and a thicker graphite coating (artists bottled graphite powder was utilized on the glass sheets for the eggplant and potato cells rather than shaved down graphite, thus creating a thicker coating before annealing).

Fig.14 All Completed Cells (Photo Taken by Finalist, 2026)

The current hypothesis is that a plant's effectiveness as a photosynthesizer depends more on the dye opacity rather than the plant's own vibrancy. While red cabbage is a more vibrant plant than eggplant, the eggplant dye was opaquer with a brownish hue, producing the best results.

Fig.15 Dye Color Comparison (Photo Taken by Finalist, 2026)

EXTENSIONS

Further Research: More cells will be made with dyes from these plants: elderberries, chokeberries, pomegranate, concord grapes, black carrots, black rice and black corn. Testing a greater multitude of dye plants will help determine whether increased amperage readings are dye-dependent or from a minor difference in cell-to-cell creation.

Planning for Longevity: To be applied to the real world, the cells must be sealed on the edges so that the dye and liquid electrolyte don't dry out. The most viable option would be through UV resin.

Pushing Forward: To maximize current and voltage output, multiple cells of the best performing dye would be wired in a parallel circuit. This circuit may be applied to small heating apparatuses, LED lights, or small fans.

REFERENCES

Ayalew, W. A., & Ayele, D. W. (2016). Dye-sensitized solar cells using natural dye as light-harvesting materials extracted from *Acanthus sennii chiovenda* flower and *Euphorbia cotinifolia* leaf. *Journal of Science: Advanced Materials and Devices*, *1*(4), 488–494. https://doi.org/10.1016/j.jsamd.2016.10.003

Is Silicone an Eco-Friendly Material? (2020, April). Takeaway Packaging; Takeaway Packaging. https://takeawaypackaging.co.uk/silicone-eco-friendly-material/#:~:text=Producing%20silicone%20uses%20hydrocarbons%20derived,it%27s%20just%20not%20the%20best]

Transformative power systems. (n.d.). Energy.Gov. Retrieved October 13, 2025, from https://www.energy.gov/fecm/transformative-power-systems